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NITI Aayog

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SECTOR AT A GLANCE

Forest-based Livelihoods

Skilling, Livelihoods and Labour Welfare Sector in India

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Skilling, Livelihoods and
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The '**Sector-at-a-Glance**' provides an easy-to-read overview and basic understanding of a sector, sub-sector, or thematic area. The facts and insights on the sector are supported by relevant data and best practices from credible government and multilateral sources. The document covers the structure, key trends, opportunities and challenges, government initiatives, and the way forward for the sector.

Suggested Citation:

NITI Aayog (2023). Forest-based Livelihoods. NITI Aayog <Sector-at-a-Glance> Series – Skilling, Livelihoods, and Labour Welfare Sector in India, 2023-Skilling, Livelihoods and Labour Welfare-SAG-07, <link>

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Overview

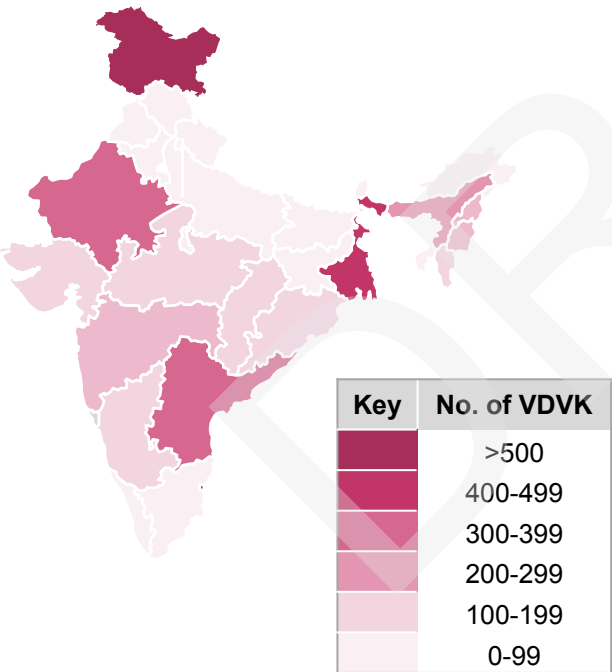
Forest-based livelihood ecosystem in India plays a pivotal role in the lives of people, particularly those residing in and around forests. It provides a wide range of services and resources that contribute significantly to the wellbeing of local communities. This ecosystem is deeply intertwined with the cultural, economic and ecological fabric of the country. In India, an estimated 73% of the forest sector workforce are informally employed.¹

India has a rich biodiversity with around **3000 Non-Timber Forest Produces (NTFP) species²** found across multiple agro-climatic zones, with almost 126 marketable species, including medicinal and edible plants, starches, gums, resins, essential oils, tendu leaves and animal products. NTFPs in India contributed to an income equivalent to **26000 Crore during 2021-22**, which accounted for 14% of the total agriculture Gross Value Added (at constant prices).³ The significance of the NTFPs in livelihood generation is being increasingly realized through better understanding of their value to the economy and environment.

The Forest Livelihood Sub-Sector is a diverse and sustainable component that contributes significantly to the well-being of communities. This sub-sector encompasses various resources and activities classified into five major categories.

- 1. Edible Resources (Wild Edible Plants, Fruits and Berries, Nuts and Seeds)
- 2. Medicinal Resources (Herbs, Roots, Bark)
- 3. Aromatic Resources (Essential Oil and Fragrances, Natural Resin & Gums from Trees)
- 4. Material Resources (Plant Fibers for Weaving, Soft materials for stuffing various products, Bamboo Shoots, Rattan for Furniture and Handicrafts, Tans and Dyes, Fuelwood)
- 5. Animal-Based Resources (Honey and Beeswax, Lac and Shellac from Insects, Silk from Forest-Dwelling Silk-Producing Insects)

Figure 1: Heat Map of India indicating VDVK clusters 6



Forest dweller communities are highly dependent on forests for their livelihoods, and this holds true from east to west as the central tropical corridor depends entirely on forest resources. Forests have been **supporting livelihoods of more than 250 million people⁴**, including millions of schedule tribe communities. This tribal dominated corridor constitutes of North-eastern states, Jharkhand, Bihar, Chhattisgarh, Madhya Pradesh, Maharashtra, Rajasthan and Gujarat. These states are home to more than 70%⁵ of India's tribal communities. Based on the growth trend, the absolute population of the tribal community was about 12.5 crores⁶ in 2020.

This **Sector at a Glance** document captures the insights and trends on three subtopics under Forest-based livelihood, namely Industrial and Plant produce, Medicine and Edibles and Worm and Bee produce.

The Ministry of Tribal Affairs and TRIFED have implemented the Van Dhan Scheme across 27 states that are rich in minor forest produce reserve. As a part of the scheme, around 3200 Van Dhan Vikas Kendras (VDVK) clusters have been established to provide Tribal Forest gatherers with economies of scale, livelihood and market-linkages.

Source: [TRIFED. \(2023\). Van Dhan Yoiana – State Wise Implementation Status. Retrieved July 15, 2023.](#)

Geographical area under forest cover in India in 2021

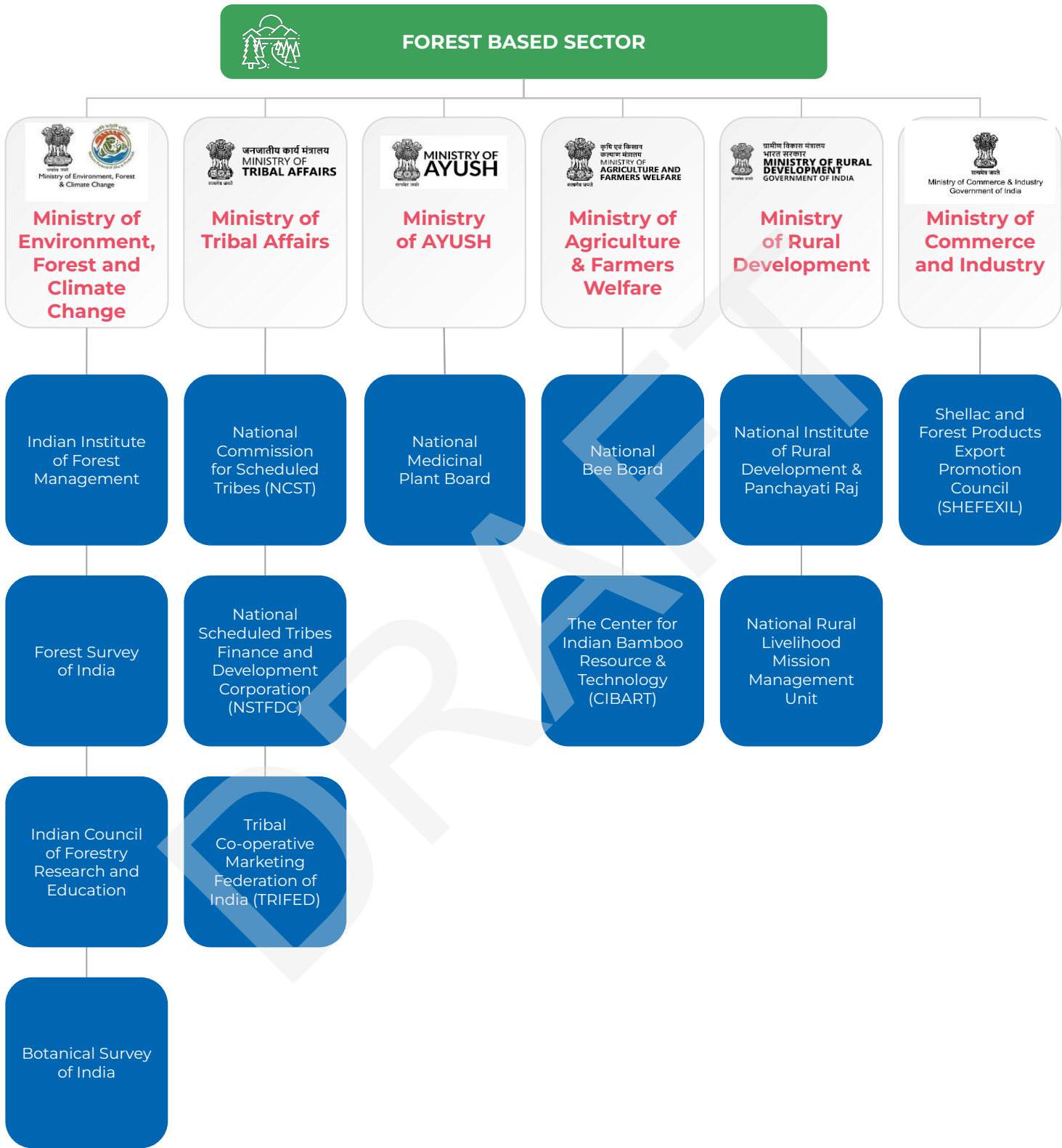
24.62%

Source: Forest Survey Report 2021

¹International labour Organization. (2022). *Contribution of the forest sector to total employment in national economies*.
²Yadav, M., & Dugava, D. (n.d.). Issues in NTFP marketing. Indira Gandhi National Forest Academy. Retrieved July 15, 2023.
³Ministry of Statistics & Programme Implementation (MoSPI). (2023). *National Accounts Statistics. Statement 8.3*.
⁴MoEF [Ministry of Environment and Forests]. 2014. National Mission for a Green India. New Delhi: Ministry of Environment and Forests, Government of India.
⁵Deutsche Gesellschaft für Internationale Zusammenarbeit (GIZ). (2022). *Expanding Trees And Forest-Based Livelihood Opportunities Through Mahatma Gandhi NREGA An Exploration*.
⁶Deutsche Gesellschaft für Internationale Zusammenarbeit (GIZ). (2022). *Expanding Trees And Forest-Based Livelihood Opportunities Through Mahatma Gandhi NREGA An Exploration*.
⁶TRIFED. (2023). *Van Dhan Yoiana – State Wise Implementation Status. Retrieved July 15, 2023.*

Major Central Institutions of The Forest-based Livelihoods Ecosystem

Figure 2: Major Government Institutions of the Forest-based Livelihoods Ecosystem



Evolution of Forest-based Livelihoods Sector

The forest-based livelihoods sector in India has undergone significant evolution over the years. Here is a brief overview:

Figure 3: Evolution of Forest-based Livelihoods Sector

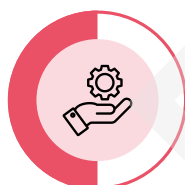
Pre-Colonial Period (Before 1857)

In India, various tribal communities had inhabited dense natural woods during the pre-colonial period. Over forest land and other resources, they had established their traditional rights. They were unrestricted in their use of the resources found in the woodlands where they resided.



Colonial Period (From 1858-1947)- II

Numerous tribal uprisings and protests colonial infringement on their rights also took place during this time. The British were compelled by the waves of forest dwellers' uprisings to pass pro-tribal laws like the Scheduled District Act, 1874, which served as the model for the Vth and VIth Schedule under Article 244, as well as local regulations like the Chhotanagpur Tenancy Act, 1908.



Colonial Period (From 1858-1947)- I

The British forest policy sought to commercially exploit the forests at the expense of the tribal peoples' secure livelihoods. Over time, the government exerted more control over the forests. In order to control people's rights to forest lands and produce, the Forest Department was established in 1864 and the Indian Forest Service was founded in 1867.



Post Colonial Period (After 1947 till Late 19th Century)

In the period from 1947 to 1987, there seemed to be an internal colonisation of the forests and tribals. The National Forest Policy, 1952 put rigid restrictions on the tribals and others on accessing timber and non-timber forest products. With the passage of Forest Conservation Act, 1980, exclusionary wildlife and forest conservation became a major concern for the government.



Conservation Era (Late 19th to Early 20th Century)

Concerns over deforestation and environmental degradation led to the emergence of the Forest Conservation Act. National parks and protected areas were established to preserve biodiversity and ecosystem services. While these measures had positive environmental impacts, they often displaced local communities that depended on these areas for their livelihoods



Modern Era (20th Century Onwards)

In recent decades, there has been a growing recognition of the importance of incorporating local communities into forest management and conservation efforts. This has led to the development of community-based forestry, agroforestry and sustainable forest management practices. International initiatives such as REDD+ (Reducing Emissions from Deforestation and Forest Degradation) have aimed to provide financial incentives for forest conservation while supporting local livelihoods.

Advances in technology, such as Remote Sensing and Geographic Information Systems (GIS) used by Forest Survey of India (FSI), have improved forest management and monitoring. This has allowed for more sustainable harvesting practices and better protection of forest ecosystems.

²Ministry of Rural Development, National Institute of Rural Development and Panchayati Raj. (2020). *Securing Forest Rights And Livelihoods of Tribals*.

Policy Evolution

Some of the milestones for GOI led initiatives in the forest-based livelihood sector are given below:

Figure 4: Glimpse of Major Policy level initiatives



Stakeholder Mapping

Figure 5: Stakeholders of Forest Livelihoods Ecosystem

Central Ministries

- Ministry of Rural Development
- Ministry of Tribal Affairs
- Ministry of Agriculture & Farmers' Welfare
- Ministry of Environment, Forest and Climate Change
- Ministry of Commerce and Industry

Implementing Bodies

- District Project Coordination Units
- Block Project Implementation Units
- Respective Dept. Support Unit
- Banks
- Training Institutes
- State Implementing Agency like SRLMs
- Technical Support Agency (TSA)
- Farmers Producer Organizations (FPOs) & NGOs
- Forest Development Agency
- Joint Forest Management Committee
- Gram Panchayats



Beneficiaries

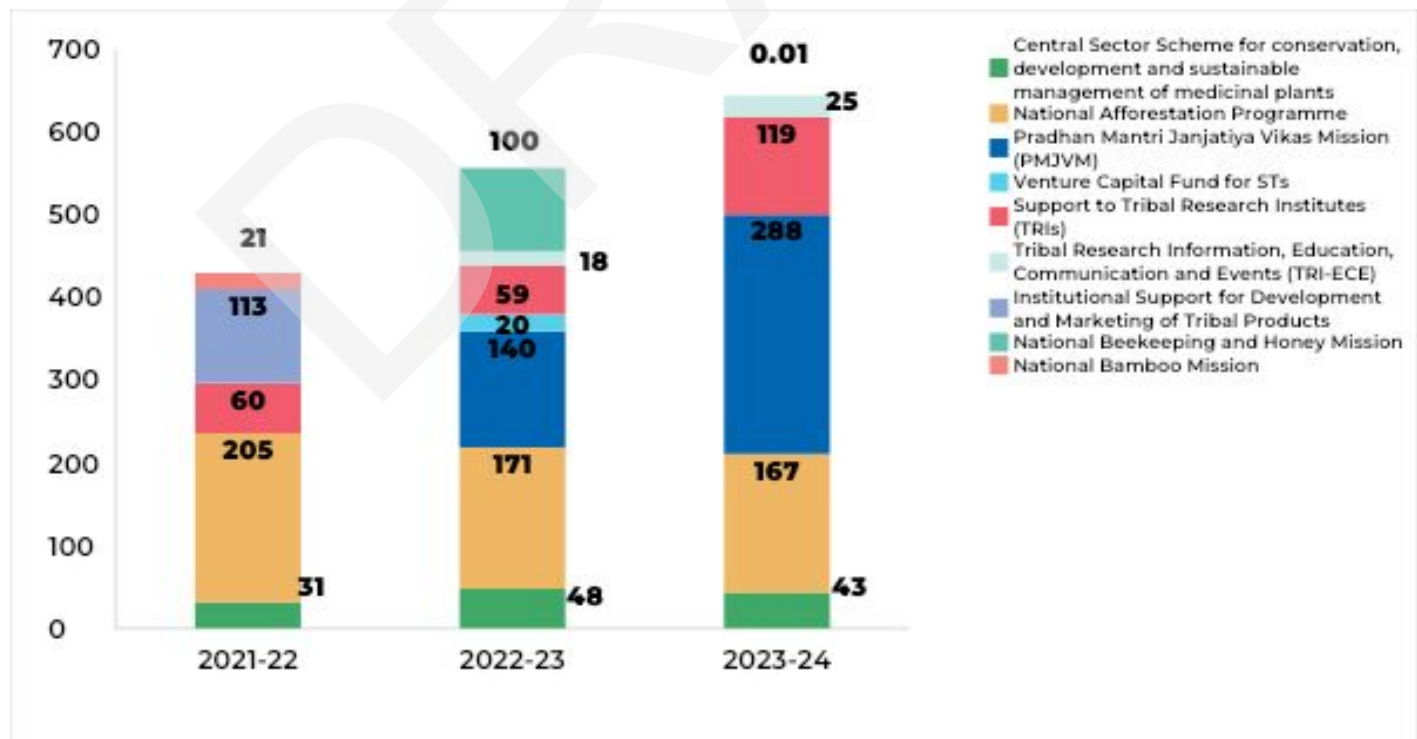
- Collectors and gatherers
- Tribal communities
- Rural Poor Households
- Self Help Groups

Enablers

- National Institute of Rural Development & Panchayati Raj (NIRD& PR)
- National Rural Livelihood Mission
- Tribal Co-Operative Marketing Development Federation of India
- Civil Society organizations
- Prime Minister's Employment Generation Programme (PMEGP)
- Shellac and Forest Products Export Promotion Council (SHEFEXIL)

Union Budget Allocation for Forest Sub-sectors

Figure 6: Scheme-wise Union Budget Allocation for forest sub-sectors (INR Cr)



Source: India Budget, 2023 and Press Information Bureau (PIB), 2023

NOTE: Includes cross-cutting schemes targeted at benefitting the rural/tribal communities involved in NTFP sector

The schemes with budget allocations available and directly impacting the forest livelihood sector as per the taxonomy have only been considered. However, there does exist umbrella schemes with partial impact on the forest livelihood sector, but bifurcation of funds allocated is missing.

Sub-sectoral Trends

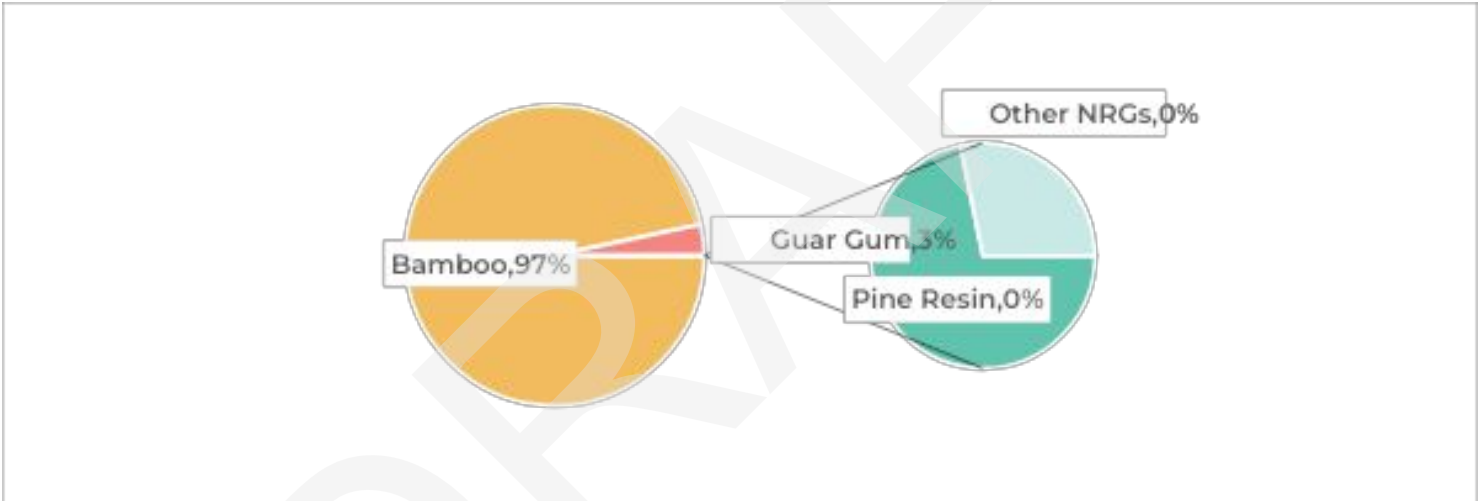
Industrial Plant Produce

Industrial Plant Produce plays a crucial role in the economic development and trade expansion of the country. Procurement and processing of industrial plant produce such as bamboo, tendu leaves, gums and resins, functions as a source of income for a significant proportion of rural and tribal population.

It is estimated that almost 2 million crafts persons in India depend on the bamboo value chain and are involved in its harvesting, processing and value addition.⁸ India is also native to guar from which the guar gum is produced. This is subsequently used extensively in the oil and gas, food and textile industries. Resins form an important segment of the industrial plant produce as they are used to produce adhesives and varnishes.⁹ Tendu leaf is another significant secondary forest produce which is collected by the forest dwellers for the production of bidis. Industrial plant produce functions not just as an essential raw material for multiple industries but also employs large sections of tribal communities.

India is one of the highest cultivator and exporter of Industrial Plant Products in the world. The below data depicts the latest production share of major Industrial plant products in a financial year.

Figure 7: Total Production share of Industrial Plant produce during 2019-20

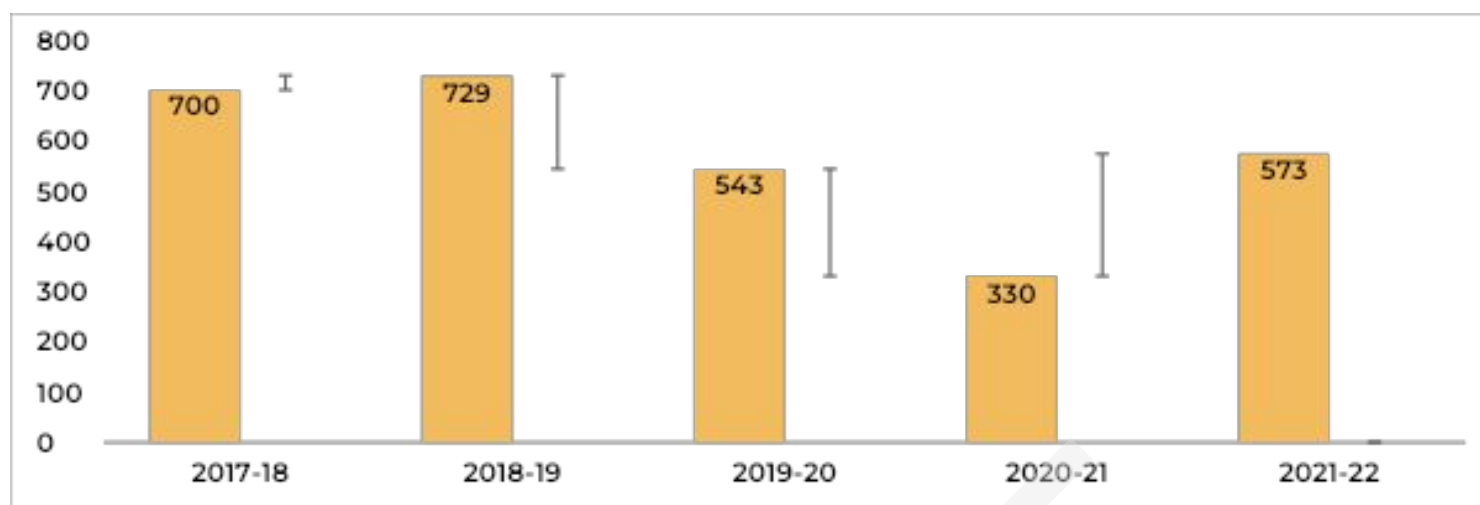


Source: Lac, Plant Resins & Gums Statistics 2020: At a Glance; Indian State of Forest Report 2021

⁸Forest Survey of India. (2021). State of Forest Report.

⁹ICAR-Indian Institute of Natural Resins and Gums, Ranchi (Jharkhand), India.(2021). Lac, Plant Resins and Gums Statistics 2019: At a Glance

Figure 8: Export value of Industrial Plant Produce from 2017-2022 (In USD Mn)



Source: Shellac and Forest Products Export Promotion Council (SHEFEXIL) Annual Reports (Multiple reports, 2017-2022)

Key Statistics



Production

- India is the **2nd largest cultivator of bamboo**, with 136 species and 23 genera spread over 13.96 million hectares¹⁰
- In 2019, Madhya Pradesh was the state with the largest area** under bamboo forests in India¹¹
- Chhattisgarh is the leading producer of the best quality Tendu leaves**, used for wrapping Beedis. The state produces approximately 16.72 lacs standard bags annually, which is **around 20% of the total Tendu leaves production of the country**¹²



Exports

- As per the year 2022-23, India is the largest exporter of guar gum** as about **90% of the world's guar gum comes from India**¹³



Employment

- Tendu leaves collection **provides employment to more than 4 million people** in India, especially the tribal population¹⁴

India has diverse and conducive agro-climatic conditions which allows the production of abundant forest-based industrial products. The growing export trends of bamboo, guar gum and other industrial plant produce is a strong testimonial of the growing demand in foreign markets. Yet, the sector is plagued by challenges such as excessive exploitation of forests, lack of information and knowledge, lack of awareness and limited access to markets, among others.

¹⁰National Bamboo Mission. (2019). Operational Guidelines of National Bamboo Mission

¹¹Forest Survey of India. (2021). State of Forest Report, Chapter 8: Bamboo Resources of the Country

¹²Chhattisgarh State Minor Forest Produce Co-operative Federation Ltd. (2023). Trade of Specified Minor Forest Produce - Tendu Leaves. Retrieved on July 15, 2023.

¹³Shellac And Forest Products Export Promotion Council. (2022). Annual Report 2021-2022.

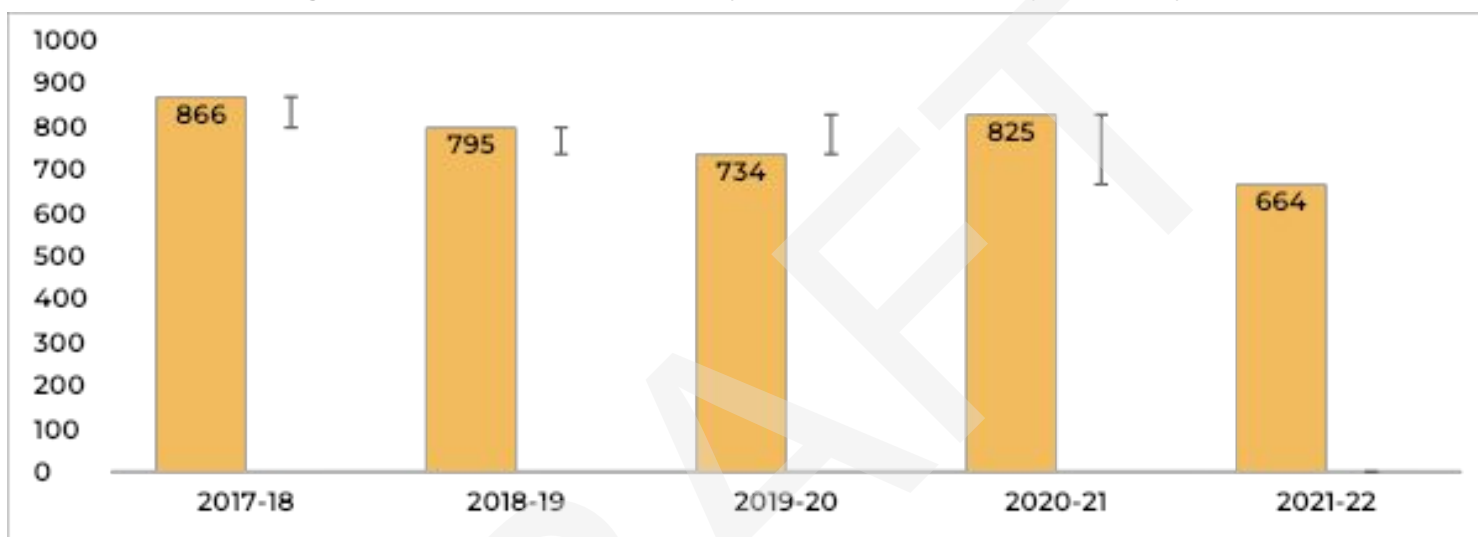
¹⁴Statista. (2022). Number of employees in tobacco industry in India as of May 2021. Retrieved on July 15, 2023.

Medicinal Plants

Medicinal plants and herbs have a rich legacy of use within India and their value is also being recognized abroad and it has been estimated that, in total over 7500 species of plants are used by several ethnic communities. Presently, medicinal plants play a very significant role in the modern economy. NTFPs account for 70% of India's forest product exports. Among these forest products, the demand for plant extracts primarily is expected to increase in the future. India has probably the richest, oldest and diverse cultural traditions in the use of medicinal plants.¹⁵

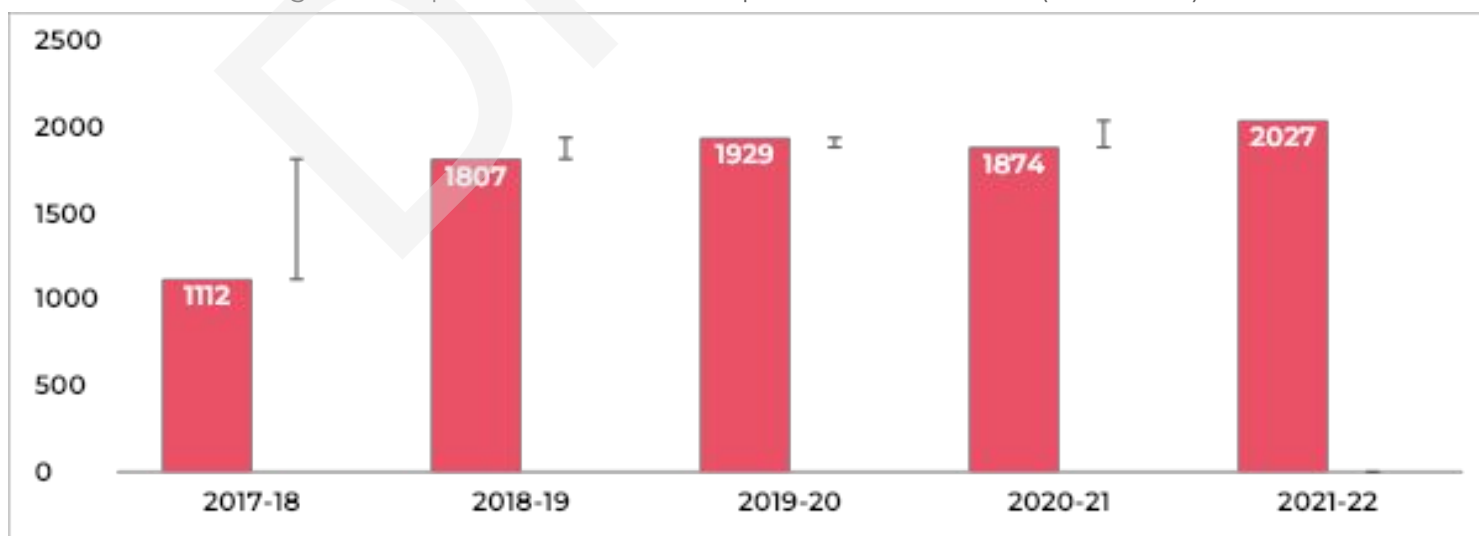
Indian forest-based plant products with medicinal properties have seen a global increase in demand, over the past years. While the demand for medicinal plants is growing, some of them are increasingly being threatened in their natural habitat. The cultivation of medicinal plant has to be encouraged for meeting the future needs and demand. Exports of plant extracts account for 42%¹⁶ of the total export value of the Shellac and Forest Products Export and Promotion.

Figure 9: Production of Medicinal plants from 2017-2022 (In '000 MT)



Source: Annual Reports of Department of Agriculture & Farmers Welfare (2018-2023)

Figure 10: Export value of Medicinal plants from 2017-2022 (In USD Mn)



Source: Shellac and Forest Products Export Promotion Council (SHEFEXIL) Annual Reports (Multiple reports, 2017-2022)

¹⁵Shellac And Forest Products Export Promotion Council. (2023). Herbs. Retrieved on July 15, 2023.

¹⁶Shellac And Forest Products Export Promotion Council. (2023). Herbs. Retrieved on July 15, 2023.

Key Statistics



Production

- In 2019, The market for medicinal plants in India was worth USD 56.6 million and is **expected to increase at a Compound Annual Growth Rate (CAGR) of 38.5%** to USD 188.6 million by 2026¹⁷
- During 2020-21, **India's total production of aromatic and medicinal crops was 825 MT**, with next year's 3rd advance estimates almost nearing this value, indicating growth in production¹⁸



Exports

- According to the International Trade Centre, Geneva, India is the 2nd largest exporter of medicinal plants¹⁹



Employment

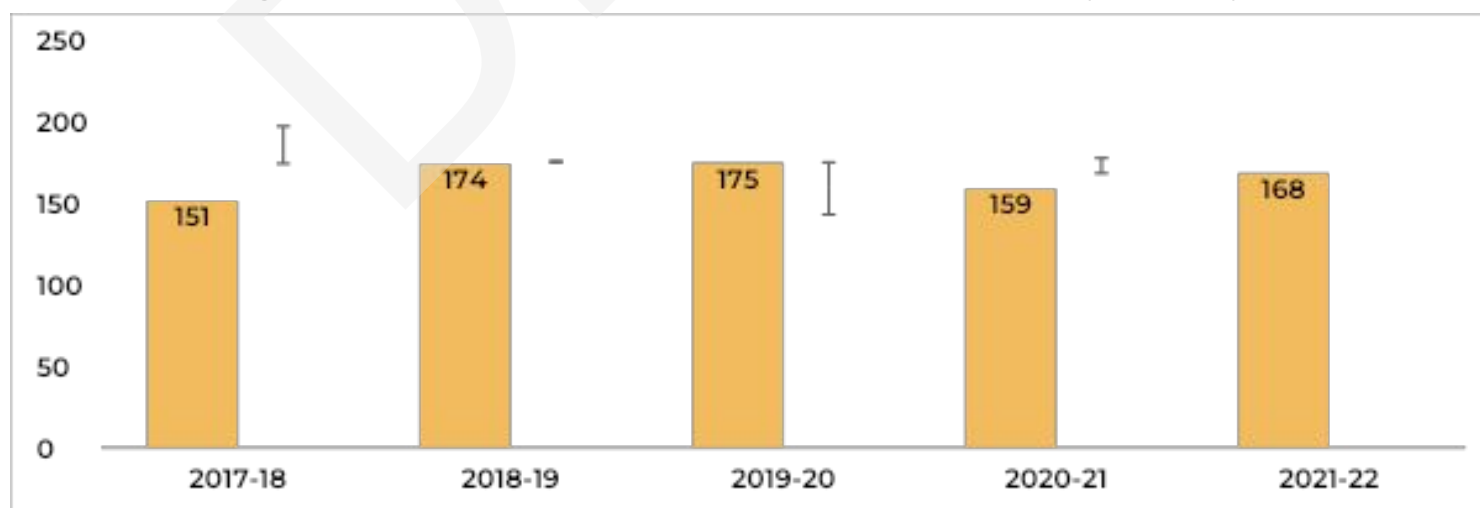
- The recent Global Ayush investment and Innovation Summit, 2022 suggested an indicative incremental employment generation of 5,35,900 jobs.²⁰

The demand for value-added medicinal plant products is gradually increasing in the global markets, yet the production and cultivation in the country remains fragmented and unorganized. A significant section of the tribal communities depends on the collection and sales of the medicinal plants for their income. Therefore, enhancing awareness among these communities and streamlining the implementation of the existing schemes would enhance the tribal incomes and further boost exports from India.

Worm and Bee produce

India holds a significant position in the world as the producer and exporter of worm and bee produces such as lac, silk and honey. With a significantly large forest cover, natural and sustainable procurement of worm and bee produce is feasible and has the potential to generate livelihood and income opportunities for the tribal and rural population. Under the scheme of 'Formation of 10,000 FPOs', total 105 FPOs have been allotted to NBB, out of which, 80 FPOs of Beekeepers/ Honey Producers have been registered/ formed till date.

Figure 11: Production of Worm and Bee Produce from 2017-2022 (In '000 MT)



Source: Annual Reports of Department of Agriculture & Farmers Welfare (2018-2023), Lac Statistics Reports of Indian Institute of Natural Resins and Gums, Ministry of Textiles Dashboard

Note: 2020-22 production data doesn't include Lac.

¹⁷Trified. (2023). Importance of Minor Forest Produces(MFPs). Retrieved on July 15, 2023.

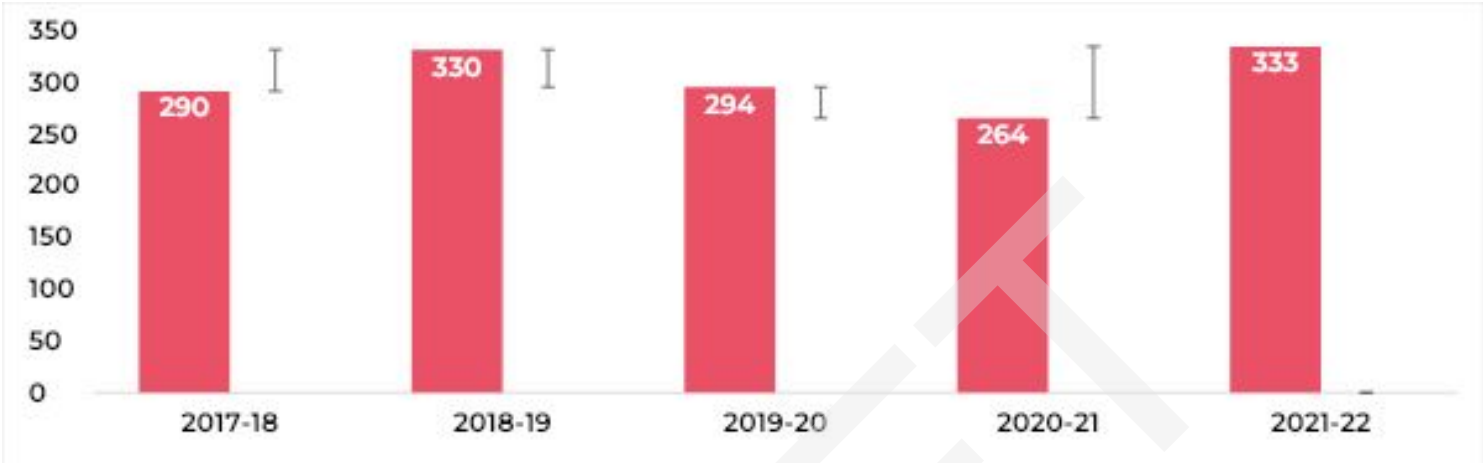
¹⁸Department of Agriculture & Farmers Welfare. (2023). Annual Report 2022-23.

¹⁹Shellac And Forest Products Export Promotion Council. (2023). Herbs. Retrieved on July 15, 2023.

²⁰Ministry of AYUSH. (2023). Annual Report (2022-2023).

Beekeeping has been recognized as an environmentally sustainable activity as it promotes nutrition, livelihood and ecological balance. Beekeepers primarily extract honey and deal in wax and pollen to some extent. India also has a blooming sericulture industry and exports natural silk as well as silk yarns and silk garments. Yet, the current production of raw silk is much lower than the present demand of silk and silk products in the country.²¹ Lac production and collection is another essential livelihood activity for the forest dwellers and sub-forest dwellers. The activity is labour intensive and can be highly remunerative for the stakeholders.²²

Figure 12: Export of Worm and Bee Produce from 2017-2022 (In USD Mn)



Source: Shellac and Forest Products Export Promotion Council (SHEFEXIL) Annual Reports (Multiple reports, 2017-2022) and The Indian Silk Export Promotion Council, 2023
Note: The data represented in the graph includes the trade values of shellac, lac and silk.

Key Statistics

Production

- India is the **2nd largest producer of silk** in the world ²³
- Around **1,26,000 households** are involved in tussar silk cultivation²⁴
- Out of the total 1,33,200 MT of honey produced in 2021-22, **55.9% was exported** to other countries²⁵

Exports

- Export of lac-based products was USD 84.44 million for the year 2021-22 which was a **28.5% increase** from the USD 65.71 million in 2020-21.
- The export earnings from silk goods was **INR 1,848.96 crore (USD 248.56 million) during 2021-22** compared to INR 1,466.60 crore (USD 198.30 million) in 2020-21.²⁶

Employment

- The estimated employment generation under sericulture in the country was **8.7 million persons during 2020-21** compared to **9.4 million persons in 2019-20**, indicating a **reduction of 7.4%**.²⁸
- 13,285** Beekeepers/Beekeeping & Honey Societies/ Firms/ Companies **with 20.05 lakhs honeybee colonies** registered with NBB and uploaded on **'Madhukranti Portal'** up to January 2023²⁹

Sericulture and beekeeping are key income generating activities in the rural and semi-urban regions of the country also both these activities have immense potential to generate foreign exchange earnings. Further, the country has registered a steady growth in the production and export of lac-based products in the last 5 years. To take the trends forward the initiatives can be undertaken to boost production of focus bee and worm-based products, strengthen the value chain and build capacity of the stakeholders.

²¹International Sericultural Commission. (2023). Statistics. Retrieved on July 15, 2023.
²²CAR-Indian Institute of Natural Resins and Gums, Ranchi (Jharkhand), India. (2021). Lac, Plant Resins and Gums Statistics 2019: At a Glance
²³International Sericultural Commission. (2023). Statistics. Retrieved on July 15, 2023.
²⁴Trifed. (2023). Importance of Minor Forest Produces(MFPs). Retrieved on July 15, 2023.
²⁵Department of Agriculture & Farmers Welfare. (2023). Annual Report 2022-23.
²⁶Shellac And Forest Products Export Promotion Council. (2022). Annual Report 2021-2022.
²⁷Central Silk Board. (2021). Annual Report 2021-22.
²⁸Central Silk Board. (2021). Functioning Of Central Silk Board & Performance Of Indian Silk Industry.
²⁹Department of Agriculture & Farmers Welfare. (2023). Annual Report 2022-23.

Opportunities and Challenges

Securing sustainable forest-based livelihoods for tribals and other traditional forest dwelling communities entail multiple complex challenges. Some of the major challenges and the key opportunities associated with them are presented in detail in the below table:

Table 1: Opportunities & Challenges in the Forest livelihood sector



Particulars	Challenges	Opportunities
Sustainable Harvesting and Collection	- Excessive exploitation of forest to procure forest produces - Destruction of forest and depletion of raw materials	The importance of forest conservation and sustainability should be explained to the collectors and other participants of the forest sector. This will help them to harvest, and extract produces judiciously through sustainable means. Progressing and sustainable forest conservation initiatives need to be encouraged. REDD+ will boost climate mitigation and offer carbon payments for forest conservation.
Social Capital and collectivization	- Fragmented ecosystem - Weak community institutions having lack of managerial and entrepreneurial capacity	Providing literacy training, and capacity-building programs on topics such as leadership, conflict resolution, communication, and governance can empower them to actively participate in collective decision-making processes.
Post-Harvest Technologies & Value Addition	- Resistance to change/adapt to new technologies and practices - Little or no value-addition at the primary level - World Health Organization estimate revealed that 80% of the people living in developing countries use wild plants to meet some of their health and nutritional needs. Therefore, less produce available for further processing.	- Infrastructure development: Focusing on improving infrastructure, such as electricity supply, internet connectivity, and transportation facilities, can create an enabling environment for technology adoption in the forest sector. - Awareness and capacity building: Promoting awareness and providing training about available technologies and their potential benefits can help overcome the knowledge gap and build confidence in technology adoption. - Value addition such as drying, cleaning at primary level will enhance the quality, shelf life and value of the produce
Innovation	Insufficient R&D in the field of NTFP, Majority stakeholders are marginalised and uneducated, with limited or no access to innovative techniques	- Promoting research and development: Encouraging investment in research and development activities specific to the forest sector can lead to the development of context-specific innovations. - Innovation challenges, competitions, and recognition programs can serve as incentives for innovation adoption. - Odisha forest department had launched 2 mobile application to notify people of the geo location of forests and plantation assets in the state.

Quality of Produce & Standardization

- Lack of information and knowledge
- Lack of standardisation
- Promoting research on the aspects of NTFP procurements and documenting new as well traditional and indigenous information.
- NTFP includes wide range of forest produces, standardisation of all the products into categories would make certifications and regulation easy.
- Export performance is strongly impacted by the quality: Enhancing standardisation in the quality through certifications can promote export.

Regulation and restrictions

- Complex and volatile regulations and restrictions in gathering of minor forest projects
- Lack of awareness about the changes among the tribal and rural population
- Simplifying and stabilizing the restrictions and regulations to enhance ease of compliance and minimize inconsistencies and confusion.
- Conducting awareness programmes on a regular basis.

Monitoring Mechanism

- Lack of formal management system
- Lack of reliable indicators. One example is of Forest cover, **as different types of forests store vastly different amount of carbon.** While one hectare of very dense **Tropical Semi-Evergreen Forest stores around 748 tonnes of CO₂ eq**, one hectare of very dense **Tropical Thorn Forest stores only 177 tonnes**, i.e., less than one-quarter.²⁹
- Establishing formal monitoring and management systems: The system must be equipped with the knowledge of the forest cover and the associated challenges This can lead to effective planning and monitoring

Access to finance

- Lack of financial literacy
- Lack of formal documents needed to get access to credit facilities
- Limited Collateral
- Promoting financial literacy and providing training on monetary management and record-keeping can enhance the financial capabilities and improve their access to finance.
- Innovative credit models, such as collateral-free loans based on cash flows, group lending and digital lending platforms, can help overcome the collateral and documentation challenges

Market Linkages

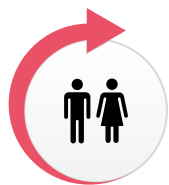
- Limited access to markets
- Price instability and fluctuations
- Supply constraints in exports
- Presence of middlemen in the supply chain Over INR 1000 crores³¹ worth of silk is exported from Indian annually. Thus, there is a huge export market for silk cloth and garments. However, the primary producers get only a fraction of the profits which are usually cornered by middlemen and trading houses.
- Improving transportation infrastructure, storage facilities, and market facilities in tribal and remote areas can enhance market access. This includes developing transportation networks, establishing collection centres, warehouses, and market yards, and promoting the use of technology for market information dissemination.
- Developing standard pricing models for minor forest products considering the supply-demand trends.
- According to the export data from Ministry of Commerce, **628 Million USD Ayush and Herbal products were exported in FY 2022-23** which has been on an increase since past years. The major importer being USA. This clearly indicates a huge potential for export of medicinal drugs.

²⁹ Forest Survey of India. (2021). State of Forest Report.

³¹ Ministry of Textiles. (2023). Textile Dashboard. Retrieved on 10th September 2023.

Cross-cutting Themes

Figure 13: Cross cutting themes in the Forest livelihood sector



Gender and Inclusivity

- **Significance**

It is primarily women who are the major gatherers of a wide range of NTFPs for subsistence and as a source of income. Yet social structures that mediate gender roles, expectations and relations pose constraints to women's active participation in local governance.

- **Policy/Programs**

Through better education, and policy interventions like Adivasi Mahila Sashaktikaran Yojana (AMSY), Joint Forest Management (JFM), gender mainstreaming and social inclusion can be achieved.

- **Way forward**

In addition to focusing on participation of women, the policies could also address the opportunity cost of participation arising from their daily livelihood, household and care responsibilities and social inequity by introducing gender specific programs for the Tribal population and other forest produce gatherers and traders.



Climate and Sustainability

- **Significance**

Forest is an integral part of the biodiversity and the environment, procurement from the forest must be done in a sustainable and climate-friendly manner. Such conscious procurements would lead to lesser depletion of and would also allow time for natural restoration which would lead to enhanced market competitiveness.

- **Policy/Programs**

National Afforestation Programme focusses on ecological restoration of degraded forests and Central Sector Scheme for conservation, development and sustainable management of medicinal plants aims to ensure sustainable procurement of medicinal plants.

- **Way forward**

Creating awareness around climate and introducing sustainable practices like sustainable procurement of resources from the forest and fairtrade.



Economic Empowerment of Tribal Communities

- **Significance**

Development of the tribal communities would ensure higher education and employment rates and will lead to greater tribal participation in economic activities.

- **Policy/Programs**

To ensure the participation of the tribal people in the country's growth process, Ministry of Tribal Affairs and TRIFED are developing and implementing policies like Pradhan Mantri Adi Adarsh Gram Yojana (PMAAGY) and Pradhan Mantri Van Dhan Yojana.

- **Way forward**

Encouraging tribal populations' inclusion and empowerment in mainstream society as well as ensuring their participation and consultation in decision-making processes.



Research & Development

- **Significance**

Promotion of research and development initiatives will encourage innovation, uncover additional forest produces and will increase awareness on tribal life and contributions.

- **Policy/Programs**

The Ministry of Tribal Affairs, under the Support to Tribal Research Institutions scheme, supports 27 Tribal Research Institutes to carry out research, documentation, and serve as a knowledge hub.

- **Way forward**

Conducting extensive research on the availability and benefits of forest produces, value addition techniques and market demand trends.



Good Governance

- **Significance**

Good governance plays a critical role in ensuring transparency, accountability, responsibility and fairness in the sector through proper monitoring of activities and implementation of policies.

- **Policy/Programs**

National Working Plan Code which was first adopted in 2004 and now has been amended in 2023 brought uniformity and will act as the guiding principle for the preparation of the working plan for scientific management of different forest divisions of our country.

- **Way forward**

Strengthening the governance of the forest-based livelihood ecosystem through monitoring mechanism and stabilising the regulations and restrictions.

Government Interventions

Table 2: Major Government Interventions in the Forest livelihood sector



KEY POLICIES, THEMES/ACTS/ PROGRAMS	MINISTRIES/ GOVERNMENT AGENCIES	TARGET GROUP	KEY FEATURES
National Forest Policy Year of Initiation – 1988	Ministry of Environment, Forest, and Climate Change	Tribal and rural population living in and around forest areas	The objectives include maintenance of environmental stability and preservation and conservation of natural heritage of India, conducting forest research and studies, checking soil erosion and desertification, and efficient utilisation of forest produce to generate livelihoods and meeting fuelwood, fodder and NTFP demands of local people.
Term Loans Year of Initiation: 2001	Ministry of Tribal Affairs	Scheduled Tribes and Other Traditional Forest Dwellers	Launched by the National Scheduled Tribes Finance and Development Corporation (NSTFDC), this scheme aims at providing term loans to beneficiaries for setting up their viable Income Generating Ventures
Adivasi Mahila Sashaktikaran Yojana (AMSY) Year of Initiation: 2002	Ministry of Tribal Affairs	Scheduled Tribe women	Launched by the National Scheduled Tribes Finance and Development Corporation (NSTFDC), this scheme aims to provide financial assistance to Tribal women
National Bamboo Mission Year of Initiation: 2006	Ministry of Agriculture and Farmers' Welfare	Bamboo cultivators and gatherers	The mission aims to promote capacity building for the bamboo sector stakeholders to ensure efficient and sustainable procurement, collection and marketing of Bamboo.
Central Sector Scheme for conservation, development and sustainable management of medicinal plants Year of Initiation: 2008	Ministry of AYUSH	Medicinal Plant harvesters	Aims at conserving degraded forests and conserve and use the Eco-Task Force mechanism for reversing habitat degradation of medicinal plants. Promotes research and development in the domain of medicinal plants and the development of agro techniques.
Adivasi Shiksha Rin Yojana (ASRY) Year of Initiation: 2011	Ministry of Tribal Affairs	Tribal Youth	Main aim of the scheme is to provide soft loans to tribal students to meet expenditure for pursuing technical and professional education
Deen Dayal Antyodaya - National Rural Livelihood Mission (DAY-NRLM) Year of Initiation: 2011	Ministry of Rural Development	SHGs, especially women SHG	In partnership with TRIFED, NRLM focusses on extending economic support programmes to ensure better livelihood opportunities, especially for women SHGs.

National Agroforestry Policy Year of Initiation: 2014	Ministry of Agriculture and Farmers' Welfare	Tribal and rural population living in and around forest areas	The main aim of the programme is ecological restoration of degraded forests and to develop the forest resources with peoples' participation, with a focus on improvement in the livelihoods of the forest-fringe communities, especially the poor.
National Beekeeping and Honey Mission Year of Initiation: 2020	Ministry of Agriculture and Farmers' Welfare	Honey Collectors	The objective is to introduce scientific beekeeping measures to the honey collectors through capacity building and trainings.
Pradhan Mantri Jan Jatiya Vikas Mission (PMJVM) Year of Initiation: 2021	Ministry of Tribal Affairs	Scheduled Tribes and Other Traditional Forest Dwellers	PMJVM has the vision to strengthen tribal entrepreneurship initiatives and to facilitate livelihood opportunities by promoting more efficient, equitable, self-managed, optimum use of natural resources, Minor Forest Produce (MFP) per farm & non-farm, tribal product-based enterprises promoted under the scheme. TRIFED is the nodal implementing agency for the scheme. PMJVM has been conceptualized with the merger of the below two schemes: Mechanism for marketing of Minor Forest Produce (MFP) through Minimum Support Price (MSP) and development of value chain for MFP: The Scheme is designed to provide fair deal to tribals by offering Minimum Support Price to the tribal MFP gatherers for forest produces collected by them. It also provisions to bridge the last mile connectivity between the tribals and markets Institutional Support for Development and Marketing of Tribal Products: The scheme supports in Retail Marketing activities such as procurement of various tribal products from the tribal suppliers, including individuals and SHGs and marketing them. Another major scheme Pradhan Mantri Van Dhan Yojana (PMVDY) has been subsumed in PMJVM. The Van Dhan Yojana encourages tribal communities to process forest produce, providing infrastructure, training, and market linkages. Implemented at village, cluster, and district levels, the scheme aims to establish 50,000 Van Dhan Vikas Kendra (VDVK) SHGs across the country, benefiting around 10 lakh tribal entrepreneurs, and establishing market linkages.
Marketing and Logistics Development for Promoting Tribal Products from the North Eastern Region (PTP-NER) Year of Approval: 2022	Ministry of Tribal Affairs	Scheduled Tribes and Other Traditional Forest Dwellers in North-Eastern Region (NER)	The scheme has been approved during 2021-22 to strengthen livelihood opportunities for tribal artisans of Northeastern States of India through increased efficiency in procurement, logistics and marketing of tribal products. TRIFED is the nodal implementing agency for the scheme.
PM- JANJATI ADIVASI NYAYA MAHA ABHIYAN (PM JANMAN) Year of Initiation: 2023	Ministry of Tribal Affairs	Particularly Vulnerable Tribal Groups (PVTGs)	The basic objective of the PM-JANMAN is to improve socio-economic conditions of the PVTGs, by saturating PVTG families and habitations with basic facilities and services. Its major objectives are: • Improvement in the status of PVTGs through skill upgradation, vocational training and other income generating opportunities • Providing basic Infrastructure in PVTG habitations and villages like road and mobile connectivity, electricity, Anganwadi facilities, education and health access.



Sustainable Practices and Green Production

- NTFP collection can be improved by conducting more research on sustainable harvesting levels and training collectors on better practices, promoting tenure security and income generation through value addition
- Projects aimed at improving the sustainable management of NTFPs need to consider the interactions between agency and social structures (which can constrain or facilitate agency) to support marginalized NTFP collectors³¹
- Introducing advanced technologies and tools to the NTFP gatherers and training them to analyse forest conditions and make livelihood decisions, based on the available knowledge



Combating Climate Change's Impact on Forest Livelihoods

- Agroforestry mitigates GHG emissions through micro-climatic modifications and carbon sequestration. As land-holding size is shrinking, tree farming combined with agriculture would optimize farm productivity and thus enhance livelihood opportunities of smallholder farmers and landless labourers, especially women³³
- Adopting a landscape-based approach to comprehensive land use planning with an emphasis on the benefits of using forest ecosystem services, such as carbon storage and sequestration, to enhance local livelihoods³⁴
- India has pledged to **sequester 2.5–3 Gt of carbon dioxide equivalent in its forests by 2030** under the Paris Agreement 2015. As per official estimates, India already has a positive carbon sequestration rate. Excluding no-go zones and densely populated areas (safeguarding livelihood of communities) would still amount to a significant constrained storage potential of 359 Gt of CO₂³⁵



Unlocking the Potential of Women Workers

- Increased focus on the empowerment of women through improved access to land, finance and training programs on sustainable forest practices, leadership and entrepreneurship development and efficient decision making
- Integration of gender lens, gender sensitisation during the trainings to make all the stakeholders understand the significance of women's role and participation
- Focus on interventions that empower women for inclusive growth as well as increasing productivity



Focusing on Research and Development to Ensure Innovation

- Increasing focus on developing research infrastructure to facilitate extensive research on the ecology and properties of the forest produces
- Documenting the traditional and indigenous knowledge of the communities to create a repository
- Encouraging research collaborations between the universities and other stakeholders to ensure innovation and evolution in the NTFP sector
- A great example for the potential of NTFPs is Bamboo, now categorized under grass, used for ethanol production, saving foreign exchange with 20% blending in petrol. Recent data indicated that Oil Marketing Companies (OMCs) had saved 433.6 crore litres of petrol on account of ethanol blending during the Ethanol Supply Year (ESY) 2021-22 resulting in an approximate savings of more than INR 20,000 crore of foreign exchange.³⁶

³¹Jalonen, R., Ziegert, R.E., Lamers, H.A.H., et al. From Within and Without: Gender, Agency and Sustainable Management of Non-Timber Forest Products in Two Indian States. *Small-scale Forestry* 22, 323–349 (2023)

³³Reorienting Indian Agriculture: Challenges and Opportunities. (2018). *Agroforestry in India* (pp. 212-218).

³⁴United States Agency for International Development. *Managing India's Forests In A Changing Climate*. (2017). *The Way Forward: A Pragmatic Strategy for Indian Forestry*.

³⁵Council on Energy, Environment and Water. (2023). *Assessing India's CO₂ Storage Potential: A Critical Analysis of What Lies Beyond the Theoretical Potential*

³⁶Ministry of Petroleum & Gas. (2023). *Ethanol Growth Story*.



Capacity Building and Knowledge Development

- Increased emphasis on enhancing the knowledge and skills of the NTFP gatherers and workers through regular and comprehensive training programs
- Introduction of region specific and product specific training programmes with high focus on the cultural and traditional needs of the targeted tribal groups
- Introducing advanced technologies and tools to the NTFP gatherers and training them to analyse forest conditions and make livelihood decisions, based on the available knowledge



Focus on optimum utilization of NTFP Sector

- Identifying and understanding the importance of NTFP as the livelihood generator for the tribal and the rural communities and its contribution to the country's economy, as a whole
- Formulating and implementing schemes to ensure productivity and processing of NTFP, promote value addition and facilitate market linkages
- Increased focus on sustainable harvesting and collection of NTFP to ensure viability of resources in the long run
- Strengthening market linkages with international customers by organizing trade fairs and connecting directly with the potential buyers. The export value for Shellac for 2021-2022 and 2022-23 remained constant at USD 106 Million, with a **YOY growth of 20% from 2020-2021**³⁶ The pre-covid numbers are 50% lower. This indicates that India has been performing well in terms of Shellac export and should continue to capture for other forest products as well.
- Developing strategies to create awareness among the consumers about the cultural significance, availability, unique qualities, benefits and environment positivity of the NTFP products to create demand and market pull for NTFP products

³⁶Ministry of Commerce and Industry. (2023). Monitoring Dashboard. Retrieved on September 10, 2023.

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